

PRACTICAL-05

AIM: Implementation of a knapsack problem using dynamic programming

PROGRAM:

```
#include<iostream>

using namespace std;

class Knapsack {
private:
    int w[10];    //weights    int
    v[10];        // values    int n;
    // number of items    int W;
    // capacity    int dp[10][100]; // dp
    table

public:
    // — Get Input —    void getInput() {    cout
    << "Enter number of items : ";    cin >> n;    cout
    << "Enter knapsack capacity : ";    cin >> W;
    for(int i = 1; i <= n; i++) {    cout << "Item " <<
    i << " weight and value : ";    cin >> w[i] >> v[i];
        }
    }

    // — Build dp table —

    void solve() {
        // base case — row 0 and col 0
        for(int i = 0; i <= n; i++) dp[i][0] = 0;
        for(int j = 0; j <= W; j++) dp[0][j] = 0;
```

```

// fill table      for(int i = 1; i
<= n; i++) {      for(int j = 1; j
<= W; j++) {

    if(w[i] <= j) {
// take or not take
dp[i][j] = max(      v[i]
+ dp[i-1][j-w[i]],
dp[i-1][j]      );
    } else {
        // cannot take item i
        dp[i][j] = dp[i-1][j];
    }
}
}
}
}

```

```

// — Print dp table — void
printTable() {      cout << "\nDP
Table:" << endl;      cout << " ";
for(int j = 0; j <= W; j++)
cout << j << " ";      cout << endl;

```

```

for(int i = 0; i <= n; i++) {
    cout << "i=" << i << " ";
for(int j = 0; j <= W; j++)
cout << dp[i][j] << " ";      cout
<< endl;
}
}

```

```
// — Find selected items — void
findItems() {    cout << "\nSelected
Items:" << endl;    int j = W;
for(int i = n; i >= 1; i--) {
if(dp[i][j] != dp[i-1][j]) {    cout
<< "Item " << i
    << " (w=" << w[i]
    << ", v=" << v[i]
<< ") TAKEN " << endl;
    j = j - w[i];    }
else {    cout << "Item
" << i
    << " (w=" << w[i]
    << ", v=" << v[i]
    << ") NOT taken " << endl;
    }
}
}
}

// — Show result — void
showResult() {    cout <<
"\nMaximum Value = "    <<
dp[n][W] << endl;
}
};

// — MAIN —
int main() {
Knapsack k;
    k.getInput();
    k.solve();
}
```

```
k.printTable();  
k.showResult();  
k.findItems();  
return 0;  
}
```

OUTPUT:

```
Output  
Enter number of items : 4  
Enter knapsack capacity : 8  
Item 1 weight and value : 2  
1  
Item 2 weight and value : 3  
2  
Item 3 weight and value : 4  
5  
Item 4 weight and value : 5  
6  
  
DP Table:  
  0  1  2  3  4  5  6  7  8  
i=0  0  0  0  0  0  0  0  0  0  
i=1  0  0  1  1  1  1  1  1  1  
i=2  0  0  1  2  2  3  3  3  3  
i=3  0  0  1  2  5  5  6  7  7  
i=4  0  0  1  2  5  6  6  7  8  
  
Maximum Value = 8  
  
Selected Items:  
Item 4 (w=5, v=6) TAKEN  
Item 3 (w=4, v=5) NOT taken  
Item 2 (w=3, v=2) TAKEN  
Item 1 (w=2, v=1) NOT taken
```

CONCLUSION:

The 0/1 Knapsack problem was successfully implemented using Dynamic Programming. The algorithm finds the maximum value within the given knapsack capacity and identifies the selected items. Its best, average and worst-case time complexity is $O(nW)$.